

THE GRAIL COLLECTOR

CFD Scientific Correction & Targeted Repair

Root cause, repaired 3-D conjugate CFD, corrected Nusselt closure, Dataset Rev 4, and the claims that can and cannot be made.

Item	Value
Decision-tree outcome	B — targeted additional CFD (executed)
3-D conjugate solutions converged	12 (was 0)
Nusselt closure	2.9238 (H2-type) → 4.48 (3-D conjugate, grid-extrapolated)
Dataset	Rev 4.1: 300 / 300; Rev 4 plus three audited derived-column corrections
Rows deleted / altered	0 — Rev 1, 2, 3 untouched and reproducible
Experimental validation	NONE claimed — all comparisons are code-to-code or code-to-reference
Roadmap V5	Gunjo et al. 2017 assessed and rejected (DESIGN_REVIEW_REQUIRED) — see Section 10

Supersedes the arrangement conclusions of the CFD Simulation Report (Rev 2) and the Dataset Audit Addendum

Contents

(In Word: right-click the field and choose Update Field.)

Summary

The production campaign's conclusions rested on a reduced-order conjugate model (GRAIL-CHT) whose Nusselt closure had never been tested against a 3-D conjugate solution, because every 3-D conjugate attempt had diverged. This work found why the 3-D solver diverged, repaired it with an exact method, produced the first converged 3-D conjugate solutions, and used them to test the model.

The model's structure passed. Its Nusselt input did not: the campaign value 2.9238 came from a boundary condition (uniform peripheral heat flux, H2) that the aluminium plate does not impose. The 3-D conjugate value is 4.48 after grid extrapolation, close to the published H1 value 4.088–4.089 for a semicircular duct. A second error was found in the model's own axial grid (110 cells, first-order): it overstated the uniformity benefit by about 3 pp.

With both corrected (Dataset Rev 4), the efficiency penalty of the alternating arrangement is larger (mean -7.3% , Mann-Whitney $p = 2e-8$, group-wise), and the plate-uniformity benefit is no longer a general result: overall the median plate standard deviation changes by only -1.1% ($p = 0.76$, not significant); it improves at high flow (-13.4% above 3.3 g/s , $p = 0.02$) and worsens at low flow ($+16.8\%$ below 2.2 g/s , $p = 0.03$). All comparisons are between the two independent 150-case groups; ALT_n and PAR_n are NOT matched pairs.

What this means for the thesis

The CFD phase is complete and defensible, but the headline claim changes from 'alternating flow improves uniformity' to 'alternating flow trades efficiency for uniformity only above a flow threshold, and costs both below it'.

This is reported because it is what the corrected physics gives. No parameter was tuned, and every decision (Nu value, grid, acceptance gates) was fixed before any Rev 4 row was seen.

1. Root-cause analysis

Every weakness is listed with its evidence and its effect. Items 1–4 are scientific; 5–7 are numerical; 8 is procedural and recorded for honesty.

Table 1 — Root causes.

#	Weakness	Evidence	Effect on results
1	Nu closure 2.9238 from an H2-type case	Fluid-only 3-D study imposed flux uniform around the perimeter; conjugate 3-D gives 4.6 (NU 64), 4.48 extrapolated; H1 reference 4.088 (Erdoğan & Imrak 2005), 4.089 (Shah & London table in Hesselgreaves 2001)	Overstated uniformity benefit ~3×; understated the efficiency penalty
2	3-D CHT divergence (chtMultiRegionSimpleFoam)	buoyant p_rgh SIMPLE fails on the fluid region alone and on the validated single-channel mesh; face fluxes correct, cell velocities wrong ~10 ⁵ in the transverse plane	No 3-D conjugate evidence existed; the ROM was unverified
3	Solar input applied per curved area in the old CHT case	solid_top 0.098723 m ² vs plan 0.088000 m ²	+12.2 % heat input (old case only; repaired with $h = U \cdot n_z$)
4	Taper has no thermal effect under constant Nu	$hP = Nu \cdot k \cdot P/Dh$ with P/Dh invariant; co-taper vs CAD-taper η identical to 5 digits	The ROM cannot represent a geometric taper benefit; any such claim is unsupported
5	ROM axial grid 110 cells, first-order	$p = 1.01$; Δstd -10.51 / -8.91 / -8.12 % at 110 / 220 / 440	~3 pp overstatement of uniformity benefit in Rev 1–3
6	3-D mass flow -0.30 % below nominal	meshed inlet 27.9726 vs CAD 28.0569 mm ²	Handled by running the ROM at the 3-D flow; negligible
7	Limit cycle at fluid relaxation 1.0	fluid minimum flipping 300.000 / 299.334 K	Removed at 0.8; steady answer unaffected
8	Operator errors in this work	empty substitution deleted a converged flow (re-run, dp 37.929 Pa reproduced); unverified mapFields -fields option (failed safely)	None on results; recorded in 14_provenance

1.1 What was NOT wrong

- The dataset's internal algebra: every derived column closes to machine precision (audit).
- The energy balances: 3-D independent balances 0.003–0.05 %; ROM < 2e-4 %.
- The ROM's structure (lumped plate, lateral conduction, channel march): with the right Nu it reproduces the co-current 3-D solution to 0.01 % in η and 0.6 % in std.
- The mesh: simpleFoam converges on the same meshes; the failure is the buoyant formulation.

2. CFD status matrix

Status: VALID — usable as evidence; QUALIFIED — usable with the stated limit; SUPERSEDED — kept for provenance, not to be cited for the conclusion it was produced for; INVALID — not to be used.

Table 2 — Every CFD product.

Product	Status	Reason / limit
Single-channel fluid-only GCI study (simpleFoam)	VALID	Hydraulics, dp, flow development
Nu = 2.9238 from that study	SUPERSEDED	H2-type; not the plate's condition
Old CHT attempts (trial2/trial3, buoyant)	INVALID	Diverged; +12.2 % solar input
3-D frozen-flow CHT, NX 60/120/240, NU 48/64/96	VALID	Converged, balances ≤ 0.05 %; exact for constant properties only
3-D CHT envelope 1.0 / 4.5 g/s (co-current)	VALID	Nu flow-independence; entry effect
NX 60 ROM-vs-3-D agreement (-8.91 vs -8.96 %)	SUPERSEDED	Coincidence of two coarse grids
Dataset Rev 1, Rev 2, Rev 3	SUPERSEDED	Nu 2.9238 and 110-cell plate grid; kept, bit-reproducible
Dataset Rev 4.1 (Rev 4 + 3 derived-column corrections)	QUALIFIED	Model-form uncertainty from the 3-D benchmark (≈ 2 pp on Δ std); campaign loss model not 3-D tested (V5 open)
Uncertainty record (efficiency -5.55 %, RMS spread -11.52 %)	SUPERSEDED	Built on Rev 2
Roadmap V5 (Gunjo et al. 2017)	REJECTED	Reference internally inconsistent; Section 10

3. Targeted CFD plan — as executed

Decision B: the evidence (a validated structure with one wrong input) called for targeted 3-D runs, not a rebuild. Every run below was completed; none was dropped.

Table 3 — Runs executed.

Purpose	Runs	Result
Repair and first solution	co + alt, NU 64, NX 60	Converged; balance 0.005 / 0.035 %
Axial grid	co + alt, NX 120, NX 240	$\Delta\eta$ converged (–18.91 %); Δ std oscillatory (–8.96 / –4.88 / –5.37 %)
Cross-section grid	co + alt, NU 48 / NR 4 and NU 96 / NR 8	Nu 4.646 / 4.602 / 4.556 $\rightarrow$ 4.48 (p 1.12, GCI 2.1 %)
Nu envelope	co, 1.0 and 4.5 g/s	Developed Nu 4.60 at all flows; entry Nu 6.1–6.35 at 4.5 g/s
ROM counterpart	Nu 4.088 / 4.48 / 4.6 / Nu(ξ); plate 110–880	p = 1.01; model-form gap 2.0 pp
Campaign	Rev 4 at plate 110 and 220	300 / 300 accepted at both

4. 3-D CHT repair

Controlled experiments, one change each, located the divergence in the buoyant solver's `p_rgh` formulation on this O-grid mesh family: it fails identically on the fluid region alone (no solid, no interface) and on the previously validated single-channel mesh. Pressure reference, wall pressure condition, bad cells, relaxation, schemes and correctors were each eliminated. The corrected face fluxes satisfy continuity while the cell velocities are wrong by $\sim 10^5$ in the cross-section plane, which is consistent with a failure in rebuilding cell velocity from face flux on cells micrometres thick and millimetres long. The source line could not be quoted, so this last step is stated as consistent with the evidence, not proven.

4.1 The repair: frozen-flow one-way coupling

With constant ρ , μ , k and c_p , the momentum and continuity equations contain no temperature. The steady flow (simpleFoam, which converges) is therefore independent of the energy solution, and solving energy on that frozen flow with `chtMultiRegionSimpleFoam`'s `frozenFlow` switch is the same steady solution as a fully coupled solve. `frozenFlow` was verified in the installed v1912 binary; 0 pressure and 0 momentum solves were confirmed; the supplied flux was confirmed unchanged (1.5e-8 relative).

Limit: the repair is exact only for constant properties. The Rev 3/4 campaign uses $k(T)$; the 3-D benchmark therefore tests the closure and the coupling, not the $k(T)$ path.

4.2 Benchmark definition

- 2-channel periodic strip, 1100×80 mm, the CAD plate (taper alternates channel to channel).
- Top: q'' 519.1256 W/m^2 minus $U_{\text{top}} 4.191176 \text{ W/m}^2\text{K} \times (T - 298.15 \text{ K})$, per unit plan area.
- Rear: $3.0 \text{ W/m}^2\text{K}$ to 298.15 K . Inlet 300 K , $2.077072\text{e-}4 \text{ kg/s}$ per channel.
- Linear loss model by design (ENGINEERING_ASSUMPTION): it isolates the coupling physics.

5. Nusselt correction

Table 4 — Conjugate Nu, developed region (15–85 % of length), co-current.

Grid / condition	Nu channel a / b	Mean
NU 64, NX 60	4.603 / 4.582	4.592
NU 64, NX 120	4.614 / 4.590	4.602
NU 64, NX 240	4.614 / 4.590	4.602
NU 48, NX 120	4.658 / 4.633	4.646
NU 96, NX 120	4.567 / 4.545	4.556
NU 64, NX 120, 1.0 g/s	4.604 / 4.595	4.599
Extrapolated (cross-section, p 1.12)	—	4.48, GCI 2.1 %
Reference H1, semicircular duct	—	4.088 / 4.089
Campaign Rev 1–3 (H2-type)	—	2.9238

Nu is axially converged and flow-independent in the developed region. It still moves with cross-section refinement, and the extrapolated value 4.48 is used for Rev 4 with a band of 4.46–4.65. The conjugate value sits above the ideal H1 value because the wall is not perfectly isothermal around the perimeter and the flat base and domed top see different fluxes.

In the alternating arrangement Nu is undefined where q' and $T_w - T_b$ change sign together ($Nu \approx 47$ at $\xi \approx 0.61$). h itself stays near 4–5 on both sides. A constant Nu therefore cannot represent the alternating channel exactly; this is the origin of the 2 pp model-form gap.

The H2 value for a semicircular duct (≈ 2.92) is recalled from Shah & London 1978 and has NOT been verified from the source. No conclusion depends on it.

6. Dataset correction — Rev 4 and Rev 4.1

Only the Nu closure and the plate grid change. Seed, Latin Hypercube, bounds, geometry gate, $k(T)$ and acceptance gates (tolerance $1e-6$, $|\text{energy error}| < 0.5\%$, $250 < T_{\text{plate}} < 500\text{ K}$) are identical to Rev 3. The outer-iteration cap was raised from 900 to 3000; the tolerance was not loosened (it is asserted in code). All 300 rows passed at both grids. No row was deleted.

Table 5 — Arrangement comparison, Dataset Rev 4.1, two independent groups (unpaired).

Quantity	Parallel (n = 150)	Alternating (n = 150)	Difference	Mann-Whitney p
η , mean	0.5620	0.5209	-7.3 %	2e-8
Plate std, median (K)	5.013	4.960	-1.1 %	0.76 (n.s.)
Plate spread, median (K)	16.57	19.61	+18.4 %	0.001
Mean plate T (K)	—	—	+13.9 K (mean)	2e-10
P90 (K)	—	—	+12.6 K (mean)	2e-6

Rev 3 (Nu 2.92, plate 110) under the same unpaired test gave η -5.1 % and plate std -29.3 % ($p = 5e-5$): the corrected closure and grid remove almost all of the apparent uniformity benefit. The input distributions of the two groups are statistically identical (KS $p = 1.00$ for all eight inputs), so the group comparison is fair. Earlier drafts of this report paired ALT_n with PAR_n; those cases do not share inputs, and every paired statistic has been withdrawn.

Table 6 — Rev 4.1 by total flow rate (terciles, unpaired).

Total flow	n alt / par	Δ mean η	Δ median plate std	p (std)
1.0 - 2.2 g/s	50 / 50	-12.7 %	+16.8 %	0.03
2.2 - 3.3 g/s	50 / 50	-6.0 %	+7.9 %	0.14 (n.s.)
3.3 - 4.5 g/s	50 / 50	-3.5 %	-13.4 %	0.02

6.1 Rev 4.1 — independent audit and surgical correction

An independent audit of Rev 4 found three derived-column groups inconsistent with their own definitions: $R4_K4 / T_R4_K / \text{mean_T_pow4_K4}$ (copied from the nx220 run into a Richardson file; 120 rows had $T_R4 < T_mean$), U_L (extrapolated linearly although it is a nonlinear function of other columns; up to 2.2 %), and Re_in / Re_out (circular-duct formula on a semicircular duct; $\times 1.3217$). Rev 4.1 recomputes only these six columns: $R4 = 2R4(220) - R4(110)$, $T_R4 = R4^{1/4}$, $\text{mean_T_pow4} = T_mean^4$, $U_L = (Q_solar - Q_u) / (0.528(T_mean - T_amb))$, $Re = \dot{m}D_h / (A\mu)$. The other 53 columns are byte-for-byte identical; no row, sign or thermal result changed. A final provenance audit found μ , Δp and energy_error_pct exact for their source (nx220) run, differing from a Richardson-consistent recomputation by at most 0.17 % and $1.9e-5$ pp — documented, not changed. Status: internally consistent within the audited model definitions and assumptions; ready to freeze.

7. Report correction plan

Table 7 — KEEP / REWRITE / QUALIFY / REMOVE.

Document / statement	Action	Replacement
CAD Model Report	KEEP	Geometry unaffected
CFD Report: hydraulics, mesh study, dp	KEEP	—
CFD Report: 'Nu = 2.9238 validated closure'	REWRITE	H2-type fluid-only value; superseded by 3-D conjugate 4.48
CFD Report: 'no conjugate 3-D solution possible'	REWRITE	Diagnosed and repaired; Sections 1, 4
CFD Report: '0 rejected' (Rev 2)	QUALIFY	True for Rev 2; Rev 3 rejected 2; Rev 4 0 at cap 3000
Audit Addendum: 'the uniformity benefit stands'	REMOVE	Flow-dependent; see Section 9
Uncertainty record headline (–5.55 % η , –11.52 % RMS)	REWRITE	Rev 4 values with model-form term
Mechanism tables (Rev 1)	REWRITE	3-D heat-recirculation budget (40 % returned)
Paper abstract: 'without simplifying assumptions'	REMOVE	Lumped plate, constant Nu, linear benchmark losses are assumptions
Paper abstract: 'closures integrated'	REWRITE	'closure derived from 3-D conjugate CFD'
Paper abstract: 'rigorous mesh independence'	QUALIFY	$\Delta\eta$ converged; Δ std oscillatory / non-asymptotic, bounded ± 1 pp
Paper abstract: mixed Nu bases across numbers	REWRITE	Every number from Rev 4
Any sentence calling a comparison 'experimental validation'	REMOVE	'benchmark against 3-D CFD' or 'comparison with published correlation'

8. Final CFD architecture

1. Level 1 — single-channel fluid-only CFD (simpleFoam, GCI): hydraulics and flow development.
2. Level 2 — 3-D conjugate CFD on the 2-channel periodic strip (simpleFoam flow, frozen-flow chtMultiRegionSimpleFoam energy), grid-studied on three axial and three cross-section levels: supplies the Nu closure and benchmarks the ROM.
3. Level 3 — GRAIL-CHT reduced-order conjugate model with the Level-2 closure (Nu 4.48), k(T), plate grid 110 and 220 with Richardson: runs the 300-case design space (Dataset Rev 4).
4. Level 4 — reference comparisons: H1 correlation (Erdoğan & Imrak 2005; Shah & London via Hesselgreaves 2001). Roadmap V5 (Gunjo et al. 2017) assessed and rejected as a reference (Section 10).

Table 8 — Uncertainty budget on the arrangement difference at design flow.

Source	$\Delta\eta$	Δ std
3-D discretisation (axial + cross-section)	± 0.1 pp	± 1.0 pp (bound)
ROM axial discretisation after Richardson	< 0.1 pp	≈ 0.4 pp
ROM model form (constant Nu, alternating)	0.5 pp (ROM less negative)	2.0 pp (ROM more negative)
Nu band 4.46–4.65	≈ 0.1 pp	≈ 1.1 pp
Entry-region Nu at high flow	< 0.1 %	≤ 2.1 % of std
Campaign loss model (glazing, radiation)	not quantified	not quantified (V5 open)

9. Final research claims

9.1 Claims supported

- Alternating counter-current flow in the converging roll-bond absorber lowers thermal efficiency: group mean -7.3% ($p = 2e-8$), at every flow tercile ($-12.7 / -6.0 / -3.5\%$). At design flow the 3-D benchmark gives -18.9% under benchmark losses.
- Mechanism: counter-flow heat recirculation through the plate — in the 3-D benchmark about 40 % of the heat a channel absorbs (9.94 of 24.80 W) is returned to the plate over 41 % of its length; the budget closes against $\dot{m}c_p\Delta T$ to 0.1 %.
- Plate uniformity depends on flow rate: alternating flow lowers the median plate std at high flow (-13.4% above 3.3 g/s, $p = 0.02$) and raises it at low flow ($+16.8\%$ below 2.2 g/s, $p = 0.03$); over the whole space the difference (-1.1%) is not significant.
- The alternating plate runs hotter on average ($+13.9\text{ K}$ mean; $+12.6\text{ K}$ mean P90 across the space).
- The 3-D conjugate Nu for the channel is 4.48 (grid-extrapolated), consistent with the H1 condition, not the H2 condition used in earlier revisions.
- A reduced-order conjugate model with this closure reproduces the 3-D co-current solution to 0.01 % in η and 0.6 % in std; for the alternating arrangement it overstates the uniformity benefit by ≈ 2 pp.

9.2 Claims that must NOT be made

- That alternating flow improves plate uniformity in general, or 'at no cost'.
- Any statistic built by pairing ALT_n with PAR_n (" 69% / 51% of pairs", " $+80\%$ / -34% "): they are independent samples.
- Any Rev 1–3 number as a final result (efficiency -4.33% , -5.24% , -5.55% ; RMS spread -11.52% ; Δ std -27.9% median).
- That the converging taper improves heat transfer: under a constant Nu the ROM cannot see it.
- That the peak-to-peak (max – min) plate temperature is reduced: group median $+18.4\%$, 3-D $+3\%$.
- 'Experimental validation' of any kind: there is none. The comparisons are 3-D-CFD-to-ROM and CFD-to-correlation.
- 'Mesh-independent' Δ std from the 3-D study: it converges oscillatorily / non-asymptotically and is bounded, not extrapolated.
- That the 3-D benchmark validates the campaign's glazing, radiation, PCM or VIP models: it uses a linear loss model and constant properties.
- Any result from the old diverging-channel geometry.

9.3 Experimental validation

None. Roadmap V5 was assessed and rejected with documented reasons (Section 10). The thesis claims verification against 3-D conjugate CFD and published laminar-duct correlations only.

Integrity statement

No row was deleted, altered or reweighted. Rev 1–3 remain bit-reproducible.

The Nu value, plate grids and gates were fixed and recorded before any Rev 4 row was seen.

Results unfavourable to the alternating arrangement are reported with the same prominence as favourable ones.

Stop point: CFD phase complete. No ANN, surrogate, optimisation or systems modelling was done.

10. Roadmap V5 — Gunjo et al. 2017 assessed

Paper: D.G. Gunjo, P. Mahanta, P.S. Robi, CFD and experimental investigation of flat plate solar water heating system under steady state condition, Renewable Energy (2017), doi:10.1016/j.renene.2016.12.041. It was read in full. Values below are read from its plots ($\pm$ about 1 K); it gives no data tables.

Table 9 — Why the reference cannot be used.

#	Finding	Evidence
1	Its simulation violates energy conservation	Noon: simulated $\Delta T \approx 33$ K needs ≈ 1.7 kW at 0.0125 kg/s and ≈ 3.4 kW at 0.025 kg/s; total solar input is ≈ 1.53 kW ($930 \text{ W/m}^2 \times 1.65 \text{ m}^2$)
2	Its accuracy claim comes from the error definition	Error taken on absolute kelvin: 336 vs 317 K is “5.7 %”, but the temperature rise is off by more than 100 %
3	Measured outlet ignores flow rate	Figs 9a/9b nearly identical at double flow; implies $\eta \approx 95$ % against the paper's own 56 % maximum
4	Inputs missing or contradictory	No wind speed, no $(\alpha\tau)$; tube wall 0.7 mm vs 7 mm; strip 0.100 m vs pitch 0.1125 m
5	Different collector	Round tubes brazed to a sheet, not a roll-bond channel plate

Decision: DESIGN_REVIEW_REQUIRED — reference rejected, V5 closed with this documented reason. No GRAIL result was compared with it, tuned to it, or described as validated by it. A replacement experimental reference, if one is required, must tabulate its data, pass an independent energy balance and state every boundary input.